



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

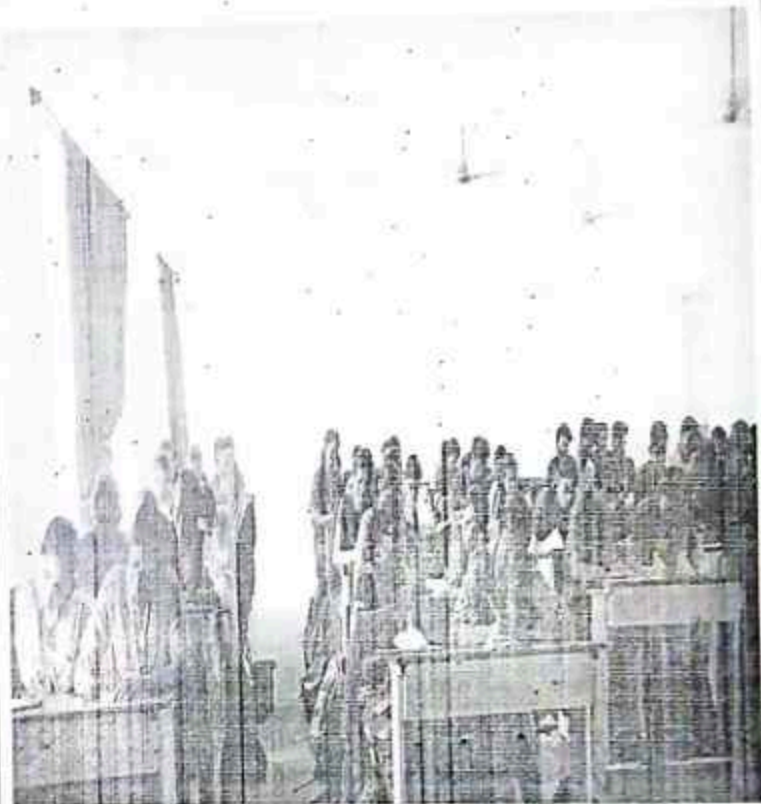
Academic Year: 2023 – 2024(Even Semester)

INNOVATIVE TEACHING METHODS

Degree, Semester & Branch : VI Semester B.E. ECE B

Course Code & Title : ET3491 & Embedded Systems and IoT Design

Name of the Faculty member: Mrs.G.Subhashini, AP/ECE

Sl.No.	Date	Topic(s)	Activity*	Reference																								
UNIT I 8051 Microcontroller																												
1.	08.02.2024	Timer and Counter	One Minute Paper	https://oncourseworkshop.com/self-awareness/one-minute-paper/																								
R. Subashini ECE PB 953601105100 Timer control registers. → TCON MSB <table border="1"><thead><tr><th>TCNT</th><th>TCOV</th><th>TCOV</th><th>TCOV</th><th>TCOV</th><th>TCOV</th><th>TCOV</th><th>TCOV</th></tr></thead><tbody><tr><td>TF1</td><td>TR1</td><td>TF0</td><td>TR0</td><td>ET1</td><td>ET0</td><td>IT1</td><td>IT0</td></tr><tr><td>8F</td><td>8E</td><td>8D</td><td>8C</td><td>8B</td><td>8A</td><td>89</td><td>88</td></tr></tbody></table> • The 8051 microcontroller has 2 timer/counters called T0 & T1. • The main purpose is to measure time & count external events. • They can be used for generating clock pulses to be used in serial communication.			TCNT	TCOV	TCOV	TCOV	TCOV	TCOV	TCOV	TCOV	TF1	TR1	TF0	TR0	ET1	ET0	IT1	IT0	8F	8E	8D	8C	8B	8A	89	88		
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TF1	TR1	TF0	TR0	ET1	ET0	IT1	IT0																					
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G. Subashini
8/2/24
Signature of the Faculty Member

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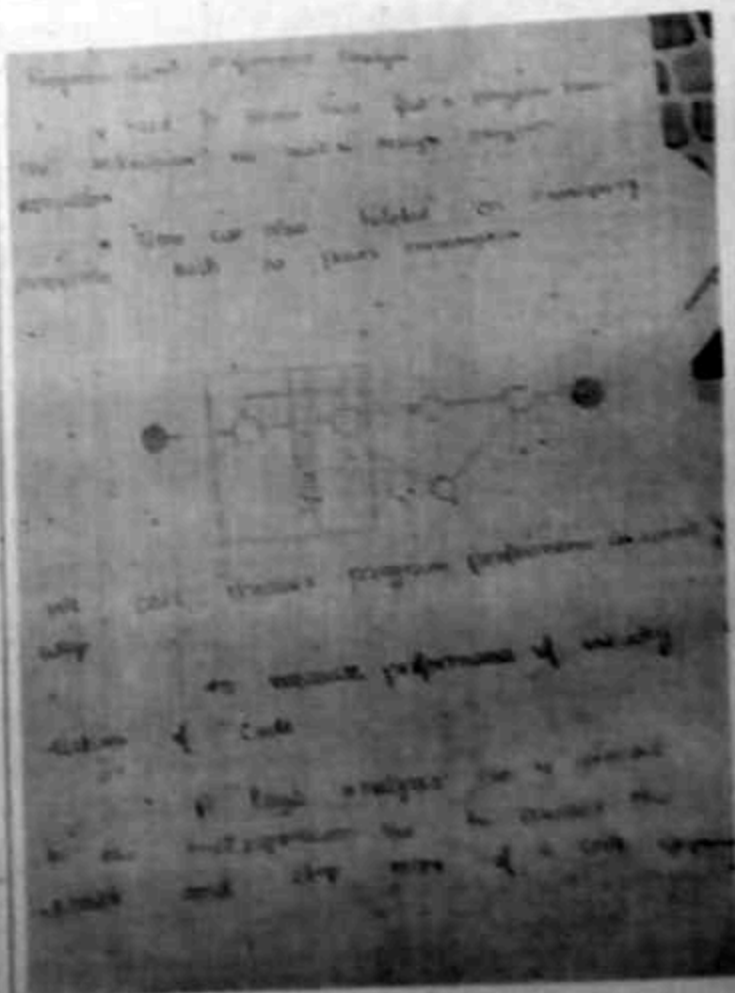
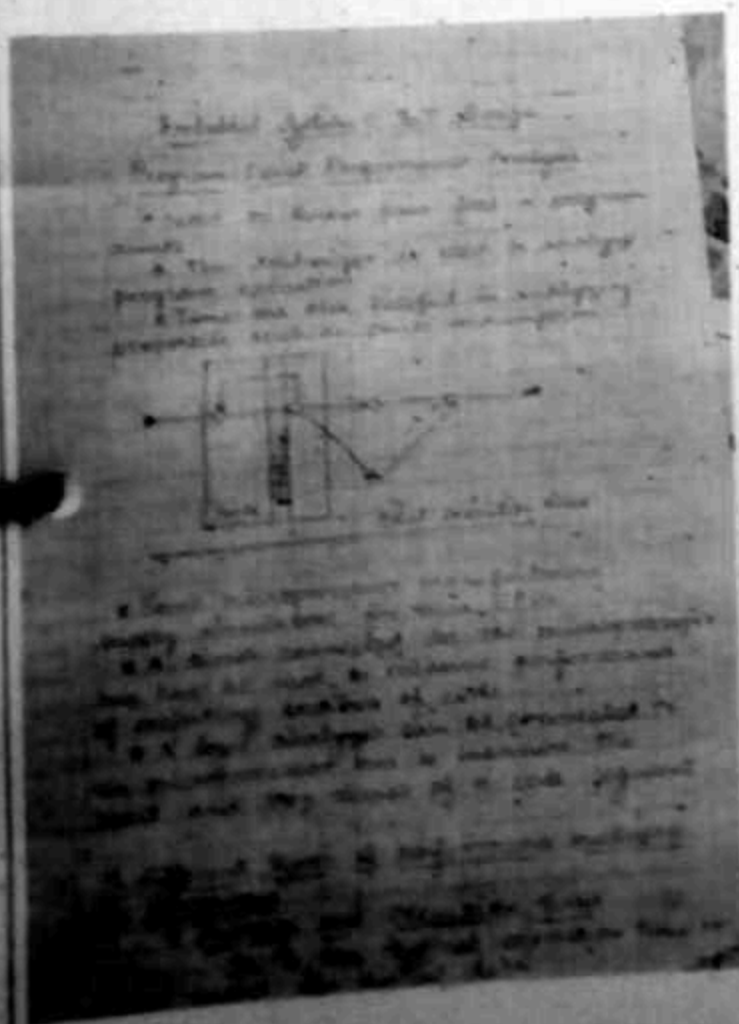
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UNIT II Embedded Systems

1.	26.02.2024	Program Level Performance Analysis.	Note Check	https://www.tisagra.com/blog/10-creative-methods-to-taking-the-note-in-classroom
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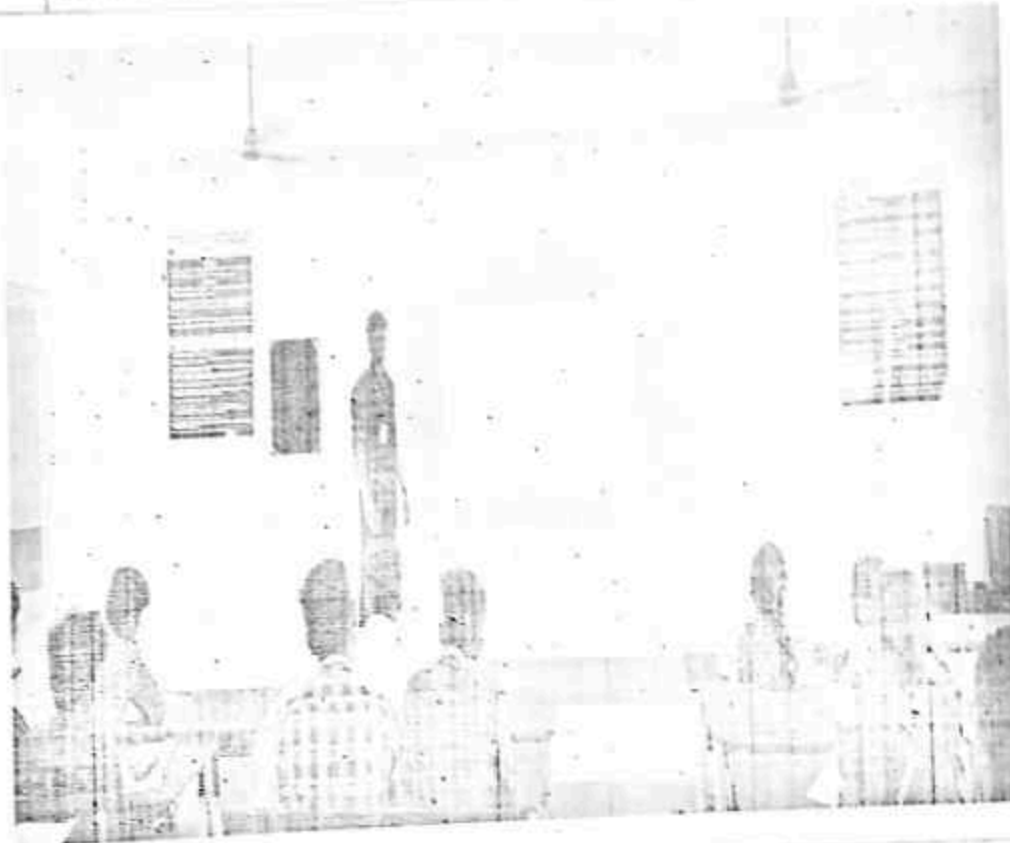
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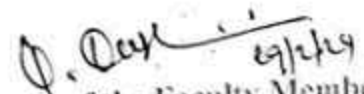
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Sl.No.	Date	Topic(s)	Activity*	Reference
UNIT III PROCESS AND OPERATING SYSTEMS				
1.	29.02.2024	Multiple tasks and Multiple Processes, Multirate Systems	One minute Speech	https://oncourseworkshop.com/self-awareness/one-minute-paper/
				


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Innovative Practice Description

- **Unit / Topic:** IV/ IoT Reference Architecture
- **Course Outcome:** CO4
- **Activity Chosen:** Flipped Class room
- **Justification:**
 - The topic of IoT Reference Architecture is chosen for the flipped classroom approach because it encompasses a comprehensive understanding of various interconnected components essential for IoT systems. This topic covers key areas such as connectivity and communications, device management, data collection, analysis, actuation, scalability, security, high availability (HA), predictive analysis, and integration. These elements are crucial for developing robust IoT solutions and align well with self-directed learning through exploration and practical application.
- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**

The overview of flipped class activity was given to students before the implementation of the activity. The students were divided into groups and one topic is allotted to each group as their wish. The topic related information and e-sources were given to the students. The students have to prepare a presentation for 10 minutes and each group has to present their topic chosen and others will listen to the presentation. After the presentation, question & answering is planned and others can share their ideas and doubts. By this way, all groups were shared their concepts learnt in their topic inside the class.

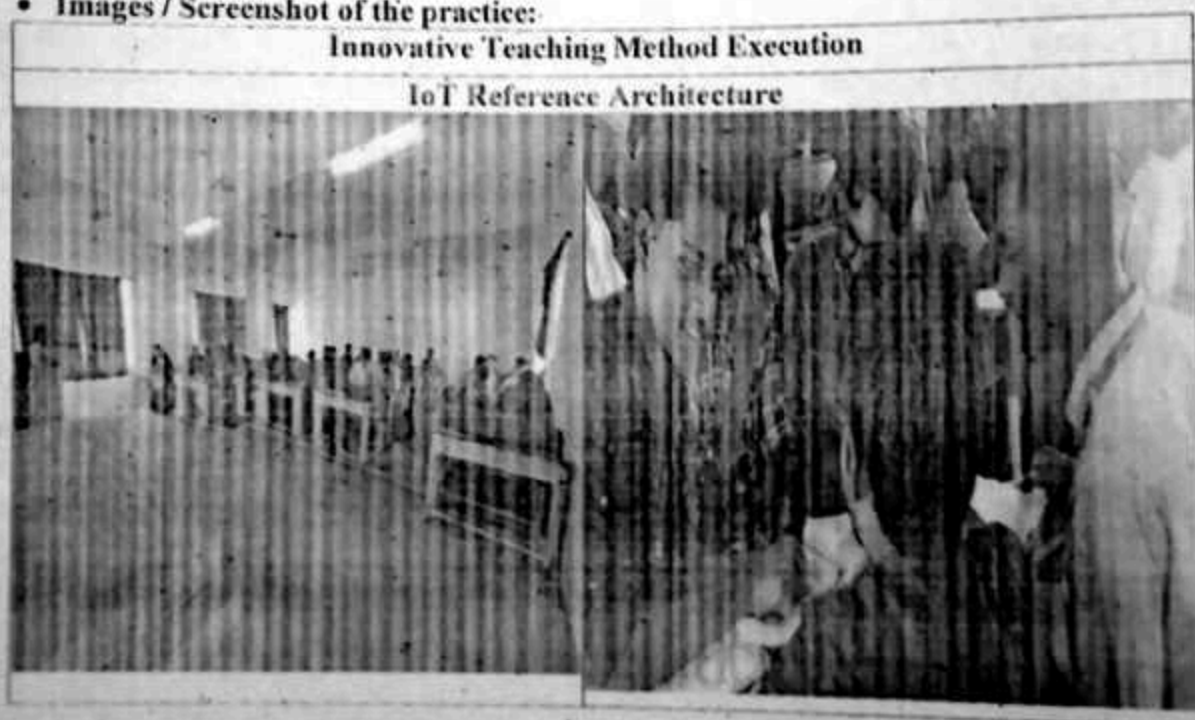
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO9
CO4	3	3	2	2	3	1	1	2

• PO / PSO mapped

Innovative practice	PSO1	PSO2	PSO4
	3	3	2
Justification for correlation	The students will be able to analyze the IoT system design process. Thus, PSO1 is mapped at level 3.	The students will be able to choose the right hardware and software for the embedded system model. Thus, PSO2 is mapped at level 3.	The students will be able to Contribute as Entrepreneurs in building Embedded products with the knowledge gained in the architecture and protocols of IoT. So, the course outcome is mapped at level 2.

• Images / Screenshot of the practice:



• Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

The students felt a new practice and it helped them to know to make good presentation and avoid stage fear. They liked this practice of teaching that can be done through this type of activity. They could get idea from learning sources which would help them to prepare the presentation.

❖ *Benefit of the practice:*

Students done this activity in a successful manner. They planned the activity for their chosen topic and prepared the presentation in the conceptual manner. All members in the group were learnt the topics thoroughly. By forming the group,

students were able to grasp the knowledge of concept by sharing the ideas among them. Each member in the group was able to present and deliver the concept of Buck, boost, buck boost analysis and design.

❖ *Challenges faced in implementation:*

Some of the students felt difficult to present because of lack of communication.

Few groups done their presentation beyond their time limit.

Few students in the group felt difficult to answer all the questions.

References:

1. Mohammed Ali Mazidi, Janice Gillispie Mazidi, Rolin D. McKinlay, The 8051 Microcontroller and Embedded Systems Using Assembly and C, Second Edition, Pearson Education, 2008.
2. Marilyn Wolf, Computers as Components – Principles of Embedded Computing System Design, Third Edition, Morgan Kaufmann, 2012.
3. Arshdeep Bahga, Vijay Madisetti, Internet of Things – A Hands on Approach, Universities Press, 2015.


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UNIT V IOT SYSTEM DESIGN				
1.	20.04.2024	Raspberry Pi – Board	Role play	https://ablconnect.harvard.edu/role-play-research



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